

Residue Theory

Complex Analysis

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Definition

Let $0 \leq r_1 < r_2$ and let $z_0 \in \mathbb{C}$. Then the **annulus around z_0 of inner radius r_1 and outer radius r_2** , denoted by $A_{r_1}^{r_2}(z_0)$, is defined to be the set

$$A_{r_1}^{r_2} = \{z \in \mathbb{C} : r_1 < |z - z_0| < r_2\}.$$

The set $A_0^r(z_0)$ is called the **punctured disc of radius r around z_0** .

Definition

A point $z_0 \in \mathbb{C}$ is called a **singularity** of a function f if f fails to be differentiable at z_0 but is differentiable at some point in every neighbourhood of z_0 . A singularity of f is said to be an **isolated singularity** if, in addition, there is a punctured disc $A_0^\varepsilon(z_0)$ around z_0 throughout which f is differentiable. An isolated singularity z_0 of f is called a

- **removable singularity** if there exists a finite $\lim_{z \rightarrow z_0} f$.
- **pole** if $\lim_{z \rightarrow z_0} f = \infty$.
- **essential singularity** if the limit $\lim_{z \rightarrow z_0} f$ does not exist.

Example

For the following functions, determine their singularities and classify them.

1. $f(z) = \frac{\sin z}{z};$

2. $f(z) = \frac{z}{1+z};$

3. $f(z) = \exp \frac{1}{z}.$

Theorem (Riemann's Principle of Removable Singularities)

A function f has a removable singularity at z_0 if and only if

$$\lim_{z \rightarrow z_0} (z - z_0)f(z) = 0.$$

Definition

Let $z_0 \in \mathbb{C}$ and let $\{c_n\}$ be a sequence in $\mathbb{C}$. An expression of the form

$$\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$$

is called a **Laurent series about** z_0 . The series is called **convergent** if both the series

$$\sum_{n=0}^{\infty} c_n (z - z_0)^n$$

and

$$\sum_{n=1}^{\infty} c_{-n} \frac{1}{(z - z_0)^n},$$

are convergent. The former and the latter are called the **regular part** (or **analytic part**) and the **principal part** of the Laurent series, respectively.

Definition

The **annulus of convergence** A of a Laurent series $\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$ is defined to be the set

$$A = \{z \in \mathbb{C} : r < |z - z_0| < R\}, \quad r = \limsup \sqrt[n]{|c_{-n}|}, \quad R = \frac{1}{\limsup \sqrt[n]{|c_n|}}.$$

The justification for the name is given in the following theorem.

Theorem

If $\{c_n\}$ is a sequence in $\mathbb{C}$ and A the annulus of convergence of a Laurent series, then the Laurent series $\sum_{n \in \mathbb{Z}} c_n (z - z_0)^n$

1. converges absolutely at each point of A .
2. diverges at each point of the set $\mathbb{C} \setminus A$.
3. can be both convergent and divergent on ∂A .
4. converges uniformly on any compact subset $K \subset A$.

Theorem

Suppose that f is analytic in the annulus $A_{r_1}^2(z_0)$. Then the representation

$$f(z) = \sum_{n \in \mathbb{Z}} a_n (z - z_0)^n,$$

called the **Laurent series of f about z_0** , is valid throughout the annulus. Furthermore, the **Laurent coefficients** are given by

$$a_n = \frac{1}{2\pi i} \int_{\gamma} d\zeta \frac{f(\zeta)}{(\zeta - z_0)^{n+1}},$$

where γ is any simple closed contour contained in the annulus that makes a complete counterclockwise revolution about the point z_0 .

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Theorem

The Laurent series of f about z_0 (in case it is valid) is unique.

Theorem (Laurent's Theorem)

Suppose $f \in \mathcal{H}(A_{\rho_1}^{\rho_2}(z_0))$, then

$$f(z) = \sum_{n \in \mathbb{Z}} c_n (z - z_0)^n \quad \text{for } z \in A_{\rho_1}^{\rho_2}(z_0).$$

Moreover, the coefficients c_n are given by

$$c_n = \frac{1}{2\pi i} \int_{\Gamma} dz \frac{f(z)}{(z - z_0)^{n+1}},$$

where Γ is the positively oriented circle of radius $\rho \in (\rho_1, \rho_2)$ and centre z_0 .

Theorem

Let $f \in \mathcal{A}(A_0^R(z_0))$ and let $z_0 \in \mathbb{C}$. The point z_0 is a pole if and only if the Laurent series of f around z_0 has no principal part, i.e.,

$$f(z) = \sum_{n=0}^{\infty} c_n (z - z_0)^n \quad \text{for all } z \in A_0^R(z_0).$$

Theorem

Let $f \in \mathcal{A}(A_0^R(z_0))$ and let $z_0 \in \mathbb{C}$. The point z_0 is a pole if and only if the principal part of the Laurent series of f around z_0 contains a finite number of nonzero terms, i.e., there exists a unique positive integer m such that

$$f(z) = \sum_{n=-m}^{\infty} c_n (z - z_0)^n \quad \text{for all } z \in A_0^R(z_0),$$

with $c_{-m} \neq 0$.

Definition

*The number m defined as in the previous theorem is called the **order** of the pole z_0 . A pole of order 1 is termed a **simple pole**.*

Theorem

Let $f \in \mathcal{A}(A_0^R(z_0))$ and let $z_0 \in \mathbb{C}$. The point z_0 is an essential singularity for the function f if and only if the principal part of the Laurent series of f around z_0 contains infinitely many nonzero terms.

Definition

Let $f \in \mathcal{A}(A_0^R(a))$, where $a \in \mathbb{C}$. The **residue of f at a** , denoted by $\text{Res}(f; a)$ (or simply, $\text{Res}_a f$) is the number

$$\text{Res}(f; a) = \frac{1}{2\pi i} \oint_{|z-a|=\rho} dz f(z) = a_{-1},$$

where $\rho \in (0, R)$ and a_{-1} is the -1 th Laurent coefficient.

Theorem (Cauchy's Residue Theorem)

Let Γ be a simple closed contour. Suppose that $f \in \mathcal{A}(\Gamma \cup \Gamma^\circ \setminus \{z_k\}_{k=1}^n)$ with $\{z_k\}_{k=1}^n \subset \Gamma^\circ$. Then

$$\oint_{\Gamma} f = 2\pi i \sum_{k=1}^n \operatorname{Res}(f; z_k).$$

Theorem

If a function f can be written as

$$f(z) = \frac{p(z)}{q(z)},$$

where p and q are analytic at z_0 , $p(z_0) \neq 0$, and q has a simple zero at z_0 , then

$$\operatorname{Res}(f; z_0) = \frac{p(z_0)}{q'(z_0)}.$$

Theorem

Suppose that z_0 is a pole of order $m \geq 1$ of f . Then we have

$$\operatorname{Res}(f; z_0) = \lim_{z \rightarrow z_0} \frac{1}{(m-1)!} \frac{d^{m-1}}{dz^{m-1}} [(z - z_0)^m f(z)].$$

Theorem

If f has a simple pole at z_0 and g is analytic at z_0 , then

$$\operatorname{Res}(fg; z_0) = g(z_0)\operatorname{Res}(f; z_0).$$

Theorem (Jordan's Lemma)

Let C_R be a semicircle of radius R in the upper half plane and centred at the origin. Let f be a function that tends uniformly to zero faster than $1/|z|$ for $\text{Arg } z \in [0, \pi]$ as $|z| \rightarrow \infty$. Let α be a nonnegative real number. Then

$$\lim_{R \rightarrow \infty} I_R := \lim_{R \rightarrow \infty} \int_{C_R} dz e^{i\alpha z} f(z) = 0.$$

Remark

So far we have discussed only integrals of functions that have no singularities on the contour. Let us now investigate the consequences of the presence of singular points on the contour. Consider the integral

$$\int_{\mathbb{R}} dx \frac{f(x)}{x - x_0},$$

where x_0 is a real number and f is analytic at x_0 . To avoid x_0 (which causes the integrand to diverge) we bypass it by indenting the contour as shown below and denoting the new contour by C_u . The contour C_0 is simply a semicircle of radius ε . For the contour C_u , we have

$$\int_{C_u} dz \frac{f(z)}{z - x_0} = \int_{-\infty}^{x_0 - \varepsilon} dx \frac{f(x)}{x - x_0} + \int_{x_0 + \varepsilon}^{\infty} dx \frac{f(x)}{x - x_0} + \int_{C_0} dz \frac{f(z)}{z - x_0}.$$

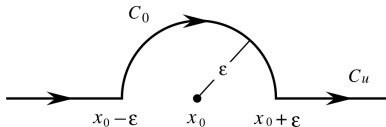


Figure 1: The contour C_u avoids x_0

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Definition

The **Cauchy's principal value** of the integral

$$\int_{\mathbb{R}} dx \frac{f(x)}{x - x_0},$$

denoted by P.V. is defined as

$$\text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = \lim_{\varepsilon \rightarrow 0} \left(\int_{-\infty}^{x_0 - \varepsilon} dx \frac{f(x)}{x - x_0} + \int_{x_0 + \varepsilon}^{\infty} dx \frac{f(x)}{x - x_0} \right)$$

Remark

The integral over the semicircle is calculated directly, yielding

$$\int_{C_0} dz \frac{f(z)}{z - x_0} = -i\pi f(x_0).$$

Therefore

$$\int_{C_u} dz \frac{f(z)}{z - x_0} = \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} - i\pi f(x_0).$$

On the other hand, if C_0 is taken below the singularity on a contour C_d , say, we obtain

$$\int_{C_d} dz \frac{f(z)}{z - x_0} = \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} + i\pi f(x_0).$$

Remark

We see that the contour integral depends on how the singular point x_0 is avoided. However, the principal value, if it exists, is unique. To calculate this principal value we close the contour by adding a large semicircle to it as before, assuming that the contribution from this semicircle goes to zero by Jordan's lemma. The contour C_u [respectively, C_d] is replaced by a closed contour, and the value of the integral will be given by the residue theorem. We therefore have

$$\text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = i\pi f(x_0) + 2\pi i \sum_j \text{Res} \left(\frac{f(z)}{z - x_0}; z_j \right),$$

$$\left[\text{respectively } \text{P.V.} \int_{\mathbb{R}} dx \frac{f(x)}{x - x_0} = -i\pi f(x_0) + 2\pi i \sum_j \text{Res} \left(\frac{f(z)}{z - x_0}; z_j \right) \right],$$

where $\{z_j\}_{j=1}^n$ are the poles of $f(z)$.